

FULL NAME: _____

BLOCK: _____

nth Roots, Rational Exponents + Cylinder DOK-3 (Item 1A Prep)

Lesson 5-1 · Quick · Student Packet

OBJECTIVES

Math Objective	Relate nth roots to rational exponents, simplify radicals with variables, evaluate rational-exponent expressions, and use nth roots to solve geometric-modeling problems.
Language Objective	Explain why $(x^{1/n})^n = x$, using radical and rational-exponent notation interchangeably.
Essential Understanding	Rational exponents ARE radicals — $x^{m/n} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$. Choose whichever form makes the computation easier.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency with nth roots and rational exponents; apply both to volume/radius modeling problems (Topic 5 assessment Item 1A prep).
Essential Question	When is rational-exponent form easier than radical form, and how do we rationalize denominators in cube-root expressions?
Materials	Student packet, pencil, calculator. DOK-3 Practice #50 (cylinder) is the lesson's capstone and rehearses the Topic 5 Assessment Item 1A move (derive variable from volume using rational exponents, then rationalize).

LAUNCH — nth Roots + Rational Exponents (teacher models Ex 1 + Ex 3)**[BOOK] nth ROOTS ↔ RATIONAL EXPONENTS (keep printed on your page)****nth ROOTS ↔ RATIONAL EXPONENTS**

1. $x^{1/n} = \sqrt[n]{x}$ (principal nth root).
2. $x^{m/n} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$.
3. To rationalize an nth-root denominator: multiply num/denom by $\sqrt[n]{x^{n-1}}$.
4. Negative nth roots: only odd n gives a real negative root; even n is undefined for negative radicands in Reals.

[WARN] COMPRESSED LESSON — TWO EXAMPLES IN LAUNCH

This is a single-period quick treatment. Teacher models BOTH Example 1 (real nth roots) AND Example 3 (evaluate rational exponents) during Launch.

Pay attention to the switch between radical form and rational-exponent form in Example 3 — that conversion is the core skill for the DOK-3 task.

EXPLORE — Think-Pair-Share (32 min)

Work with a partner. Move in order. Practice #50 (Cylinder Modeling) is the big one — plan ~10 min for it.

Bank-item labels (in order):

1. Try It 1 — Real nth roots
2. Try It 3 — Evaluate rational exponents
3. Practice #28 — nth roots with variables
4. Practice #30 — Rational exponent form
5. Practice #37 — Simplify: mixed radical/rational
6. Practice #48 — Multi-step rational exponent
7. Practice #50 [STAR] DOK-3 CYLINDER MODELING — Topic 5 Assessment Item 1A prep

Try It 1 — Real nth roots

Find the specified roots of each number.

- a. real fourth roots of 81
- b. real cube roots of 64

Try It 3 — Evaluate rational exponents

What is the value of each expression? Round to the nearest hundredth if necessary.

- a. $-(16^{(3)/(4)})$
- b. $\sqrt[5]{3.5^4}$

Practice #28 — nth roots with variables

Find the specified roots of each number.

The real square roots of 25.

Practice #30 — Rational exponent form

Rewrite each expression using a fractional exponent.

$$\sqrt[6]{729}$$

Practice #37 — Simplify: mixed radical/rational

Simplify each expression. Assume all variables are positive.

$$\sqrt[6]{729a^{24}b^{18}}$$

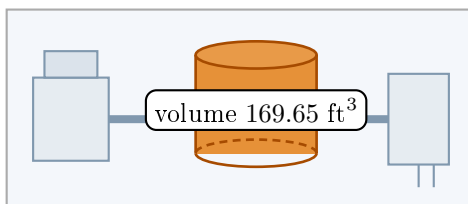
Practice #48 — Multi-step rational exponent

Determine if each expression is another way to write $b^{\frac{3}{4}}$. Select Yes or No.

Expression	Yes	No
a. $\sqrt[4]{b^3}$	☐	☐
b. $(b^3)^{\frac{1}{4}}$	☐	☐
c. $b^{\frac{4}{3}}$	☐	☐
d. $\sqrt[3]{b^4}$	☐	☐
e. $\frac{b^3}{b^4}$	☐	☐

Practice #50 [STAR] DOK-3 CYLINDER MODELING — Topic 5 Assessment Item 1A prep

Performance Task A milk processing company uses cylindrical-shaped containers. The height of the container is equal to the diameter of the base.



[IMAGE: Milk processing machine with an orange cylindrical container; label reads “volume 169.65 ft³.”]

Part A How much material is needed to make the lateral surface area of the shipping container?

Part B The cargo hold of a ship is 20 ft high. What is the largest number of these shipping containers that could be stacked on top of each other inside the cargo hold?

SHARE / SUMMARY (5 min)

Sentence frames

Pair share (Cylinder): “I isolated r from $V = \pi r^2 h$ by substituting $h = 2r$, giving $V = \underline{\hspace{1cm}}$. Solving for r , I got $r = \underline{\hspace{1cm}}$. In rational-exponent form, $r = (V/(2\pi))^{1/3}$. To rationalize I multiplied by $\underline{\hspace{1cm}}$ to get $\underline{\hspace{1cm}}$.”

Whole class: one pair reads their cylinder solution. Class signals thumbs up/sideways.

[SEARCH] BRIDGE TO TOPIC 5 ASSESSMENT

The cylinder move — isolate a variable, convert to rational-exponent form, rationalize — is exactly what the Topic 5 Assessment Item 1A asks. You have now rehearsed it.

EXIT

$x^{m/n}$ is the same as _____ in radical form.

To rationalize a cube-root denominator I _____.

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #49 is an SAT/ACT cube-root item that extends rational-exponent fluency with large radicands.

Optional #1 (Savvas Practice #49 — SAT/ACT $\sqrt[3]{4,096x^{18}y^{30}}$)

SAT/ACT Which of the following is equivalent to $\sqrt[6]{4,096x^{18}y^{30}}$, where $x > 0$ and $y > 0$?

- A. $682.7x^{15}y^{24}$
- B. $4x^{1.6}y^{1.8}$
- C. $4,096x^3y^5$
- D. $4x^3y^5$
- E. $682.7x^3y^5$